

CYBERSECURITY LABORATORY PROJECT

NMAP NETWORK RECONNAISSANCE & SERVICE EXPOSURE ASSESSMENT

Authorized Proxmox-Based Security Laboratory

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PORTFOLIO PROJECT WORKBOOK

Build → Test → Capture Evidence → Analyze → Document

01. Project Overview

PROJECT PURPOSE

Build an isolated Proxmox cybersecurity laboratory where Nmap is used to perform authorized network discovery, service/version enumeration, OS identification, and security-focused service exposure assessment. The final output should be a reproducible technical project with evidence suitable for GitHub and a cybersecurity portfolio.

Project Type	Authorized cybersecurity laboratory / defensive assessment
Primary Tool	Nmap
Virtualization	Proxmox VE
Initial Target	Ubuntu Server virtual machine
Security Scope	Only the isolated laboratory network and its intentionally created VMs
Portfolio Use	PCPR evidence + GitHub project + technical interview discussion

Project Objectives

- Create an isolated Proxmox laboratory network.
- Deploy at least one Ubuntu Server target VM and one security/scanning VM.
- Verify that the lab hosts can communicate only within the intended lab network.
- Perform network discovery and host enumeration.
- Perform service and version enumeration against authorized targets.
- Perform OS/platform identification where technically possible.
- Assess exposed services and identify security-relevant observations.
- Capture reproducible evidence for every major phase.
- Produce a professional README and findings report.
- Document limitations, safety boundaries, and lessons learned.

IMPORTANT SAFETY BOUNDARY

Do not scan public IPs, school/company networks, neighbors' devices, or any system you do not explicitly own or have authorization to assess. All scans in this workbook are intended for the isolated Proxmox lab only.

02. Final Deliverables & Folder Structure

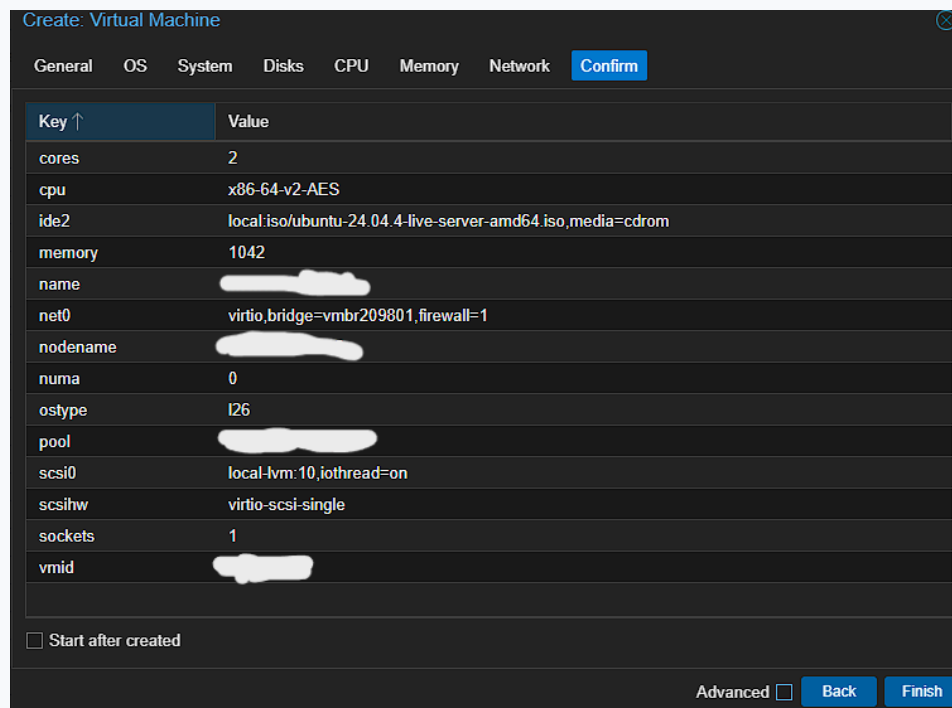
Before building the lab, create a GitHub repository. Use the structure below so the project remains organized as evidence grows.

FOLDER / FILE	PURPOSE
README.md	Project overview, scope, topology, methodology, findings summary
01-Network-Discovery/	Discovery commands, raw outputs, screenshots, analysis
02-Service-Version-Enumeration/	Service/version scan outputs and interpretation
03-OS-Identification/	OS detection outputs and limitations
04-Service-Exposure-Assessment/	Security-focused analysis of exposed services
05-Documentation/	Topology, screenshots, methodology, report
evidence/	Selected clean screenshots for portfolio use
results/	Sanitized text/XML/Nmap outputs

03. Phase 1 — Build the Proxmox Laboratory

Step 1 — Create the Ubuntu Server Target VM

EVIDENCE 01 — Proxmox Target VM



The screenshot shows the 'Create: Virtual Machine' window in Proxmox. The 'General' tab is selected. The configuration table is as follows:

Key	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/ubuntu-24.04.4-live-server-amd64.iso,media=cdrom
memory	1042
name	[REDACTED]
net0	virtio,bridge=vbr209801,firewall=1
nodename	[REDACTED]
numa	0
ostype	l26
pool	[REDACTED]
scsi0	local-lvm:10,iotread-on
scsihw	virtio-scsi-single
sockets	1
vmid	[REDACTED]

At the bottom, there is a checkbox for 'Start after created' which is unchecked. The 'Confirm' button is highlighted in blue. At the bottom right, there are 'Advanced' (unchecked), 'Back', and 'Finish' buttons.

Figure 1.1. Proxmox configuration view of the authorized Ubuntu Server target VM used for the Nmap network reconnaissance laboratory.

Step 2 — Create the Security/Scanning VM

EVIDENCE 02 — Security VM

Paste a screenshot showing the second VM running and connected to the same isolated lab network.

The screenshot shows the 'Create: Virtual Machine' window in Proxmox. The 'General' tab is selected. The configuration table is as follows:

Key ↑	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/kali-linux-2025.3-installer-amd64.iso,media=cdrom
memory	2048
name	[REDACTED]
net0	virtio,bridge=vbr209801,firewall=1
nodename	[REDACTED]
numa	0
ostype	l26
pool	[REDACTED]
scsi0	local-lvm:32,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	[REDACTED]

At the bottom, there is a checkbox for 'Start after created' which is unchecked. On the right, there are buttons for 'Advanced' (with a dropdown arrow), 'Back', and 'Finish'.

Figure 2.1. Running state of the Kali Linux security assessment VM within the Proxmox laboratory environment, used as the authorized Nmap scanning host for the network reconnaissance assessment.

04. Phase 2 — Verify the Lab Network

The purpose of this phase is to prove that the environment is working before any assessment begins.

Step 1 — Record the Lab Addresses

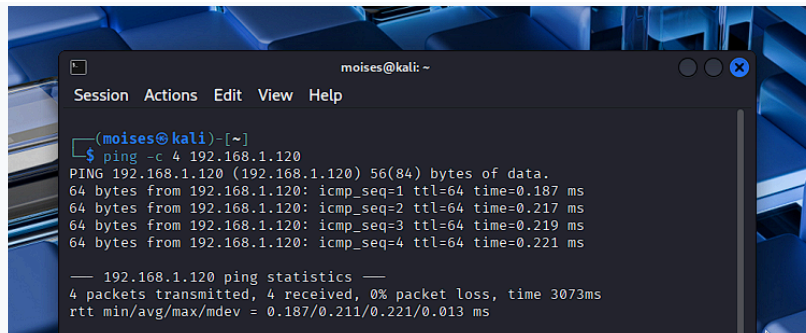
SYSTEM	HOSTNAME	LAB IP ADDRESS
Security / Scanner VM	kali	192.168.1.139/24
Ubuntu Server Target	ubuntuserver	192.168.1.120/24

Step 2 — Confirm Connectivity

From the security VM, test connectivity to the Ubuntu Server target using the target's lab IP address. Capture the successful result.

EVIDENCE 03 — Connectivity Verification

Paste the terminal screenshot showing the scanner VM successfully reaching the target VM.



```

moises@kali: ~
$ ping -c 4 192.168.1.120
PING 192.168.1.120 (192.168.1.120) 56(84) bytes of data:
64 bytes from 192.168.1.120: icmp_seq=1 ttl=64 time=0.187 ms
64 bytes from 192.168.1.120: icmp_seq=2 ttl=64 time=0.217 ms
64 bytes from 192.168.1.120: icmp_seq=3 ttl=64 time=0.219 ms
64 bytes from 192.168.1.120: icmp_seq=4 ttl=64 time=0.221 ms

--- 192.168.1.120 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3073ms
rtt min/avg/max/mdev = 0.187/0.211/0.221/0.013 ms
  
```

Figure 3.1. Connectivity verification between the Kali Linux security assessment VM and the authorized Ubuntu Server target, confirming successful network communication before Nmap reconnaissance activities.

Step 3 — Confirm the Network Boundary

This laboratory was conducted within a school-provided Proxmox cloud environment. The underlying network and bridge infrastructure is managed by the school and was not modified as part of this project. The screenshots below document the assigned laboratory virtual machines and their configured network devices.

EVIDENCE 04 — Lab Network Boundary

Summary

Add

Remove

Edit

Disk Action

Revert

> Console

Hardware

Cloud-Init

Options

Task History

Monitor

Backup

Replication

Snapshots

Memory

Processors

BIOS

Display

Machine

SCSI Controller

CD/DVD Drive (ide2)

Hard Disk (scsi0)

Network Device (net0)

2.00 GiB

2 (1 sockets, 2 cores) [x86-64-v2-AES]

Default (SeaBIOS)

Default

Default (i440fx)

VirtIO SCSI single

local-iso:kali-linux-2025.3-installer-amd64.iso,media=cdrom,size=4489380K

local-lvm:vm-100-disk-0,iothread=1,size=32G

virtio:vmbr209801,bridge=vmbz209801,firewall=1

Summary

Add

Remove

Edit

Disk Action

Revert

> Console

Hardware

Cloud-Init

Options

Task History

Monitor

Backup

Replication

Snapshots

Memory

Processors

BIOS

Display

Machine

SCSI Controller

CD/DVD Drive (ide2)

Hard Disk (scsi0)

Network Device (net0)

1.02 GiB

2 (1 sockets, 2 cores) [x86-64-v2-AES]

Default (SeaBIOS)

Default

Default (i440fx)

VirtIO SCSI single

local-iso:ubuntu-24.04.4-live-server-amd64.iso,media=cdrom,size=3325654K

local-lvm:vm-101-disk-0,iothread=1,size=10G

virtio:vmbr209801,bridge=vmbz209801,firewall=1

Figure 4.1. Assigned virtual machine network configuration within the school-provided Proxmox laboratory environment. The underlying Proxmox network infrastructure was not modified as part of this project.

05. Phase 3 — Network Discovery & Host Enumeration

Goal: identify which hosts are reachable within the authorized lab network.

Step 1 — Identify the Lab Subnet

On the scanner VM, determine the IP address and subnet assigned to the lab interface. Record the subnet below.

LAB SUBNET

192.168.1.0/24

Step 2 — Perform Host Discovery

Use Nmap host discovery against ONLY the isolated lab subnet. Record the exact command you used.

COMMAND USED

```
nmap -sn 192.168.1.120
```

Replace the bracketed text with your actual command. Do not copy a command blindly; document what you actually executed.

EVIDENCE 05 — Host Discovery Command + Result

Paste a terminal screenshot where both the command and the resulting discovered host(s) are readable.

```
nmap -sn 192.168.1.120
```

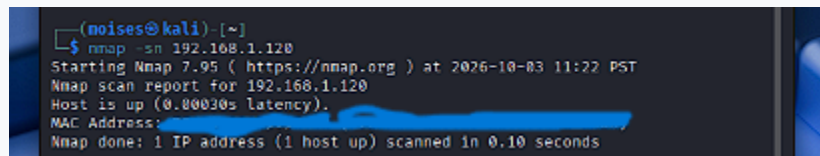


Figure 5.1. Nmap host discovery performed against the authorized Ubuntu Server target (192.168.1.120) from the security/scanning VM. The result confirms that the target host was reachable within the school-provided laboratory environment.

Step 3 — Record the Results

HOST	IP ADDRESS	STATUS	NOTES
ubuntuserver	192.168.1.120	Host is up	Authorized laboratory target

Analysis

The Nmap host discovery scan identified one reachable host at 192.168.1.120, corresponding to the Ubuntu Server target assigned for this laboratory. The result was consistent with the intended two-VM laboratory setup, where the security/scanning VM communicates with the Ubuntu Server target. No additional hosts were assessed because the scan was intentionally limited to the authorized target.

06. Phase 4 — Service & Version Enumeration

Goal: identify open ports, exposed services, and service/version information on the authorized Ubuntu Server target.

Step 1 — Target Confirmation

AUTHORIZED TARGET

Target hostname: ubuntuserver Target lab IP: 192.168.1.120

Step 2 — Run Service/Version Enumeration

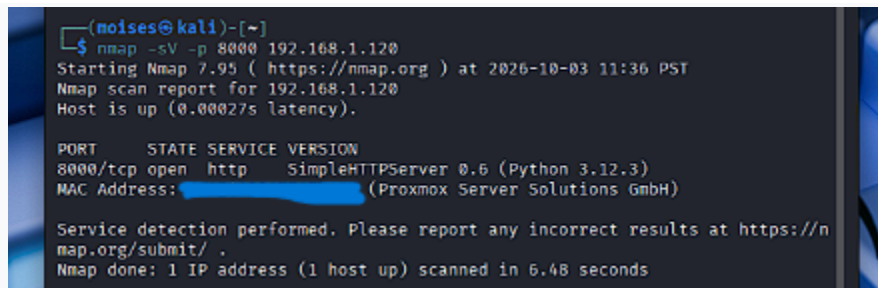
Perform service/version enumeration against the target VM. Record the exact command used and preserve the raw output in the GitHub results/ folder.

COMMAND USED

`nmap -sV -p 8000 192.168.1.120`

EVIDENCE 06 — Service & Version Enumeration

Paste the cleanest screenshot showing the Nmap command and the detected open ports/services. Crop unrelated terminal content.



```
(noises@kali)-[~]
└─$ nmap -sV -p 8000 192.168.1.120
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-03 11:36 PST
Nmap scan report for 192.168.1.120
Host is up (0.00027s latency).

PORT      STATE SERVICE VERSION
8000/tcp  open  http    SimpleHTTPServer 0.6 (Python 3.12.3)
MAC Address: XXXXXXXXXX (Proxmox Server Solutions GmbH)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 6.48 seconds
```

Figure 6.1. Nmap service and version enumeration of the authorized Ubuntu Server target, identifying TCP port 8000 as an HTTP service running SimpleHTTPServer 0.6 on Python 3.12.3. The service was intentionally enabled within the controlled laboratory environment for detection and assessment.

Step 3 — Build the Service Inventory

PORT	STATE	SERVICE	VERSION / PRODUCT	SECURITY RELEVANCE
8000/tcp	open	http	SimpleHTTPServer 0.6 (Python 3.12.3)	Intentionally exposed laboratory HTTP service; demonstrates detectable network service exposure

Analysis

The initial service enumeration did not identify an open service among Nmap's default 1,000 TCP ports. A temporary HTTP service was then intentionally enabled on TCP port 8000 within the laboratory target. A targeted Nmap service/version scan successfully identified the port as an HTTP service running SimpleHTTPServer 0.6 on

Python 3.12.3. This demonstrates how service exposure can be identified and documented through network reconnaissance.

07. Phase 5 — OS / Platform Identification

Goal: evaluate whether Nmap can identify the target operating system or platform characteristics, while documenting uncertainty.

Step 1 — Run OS Identification

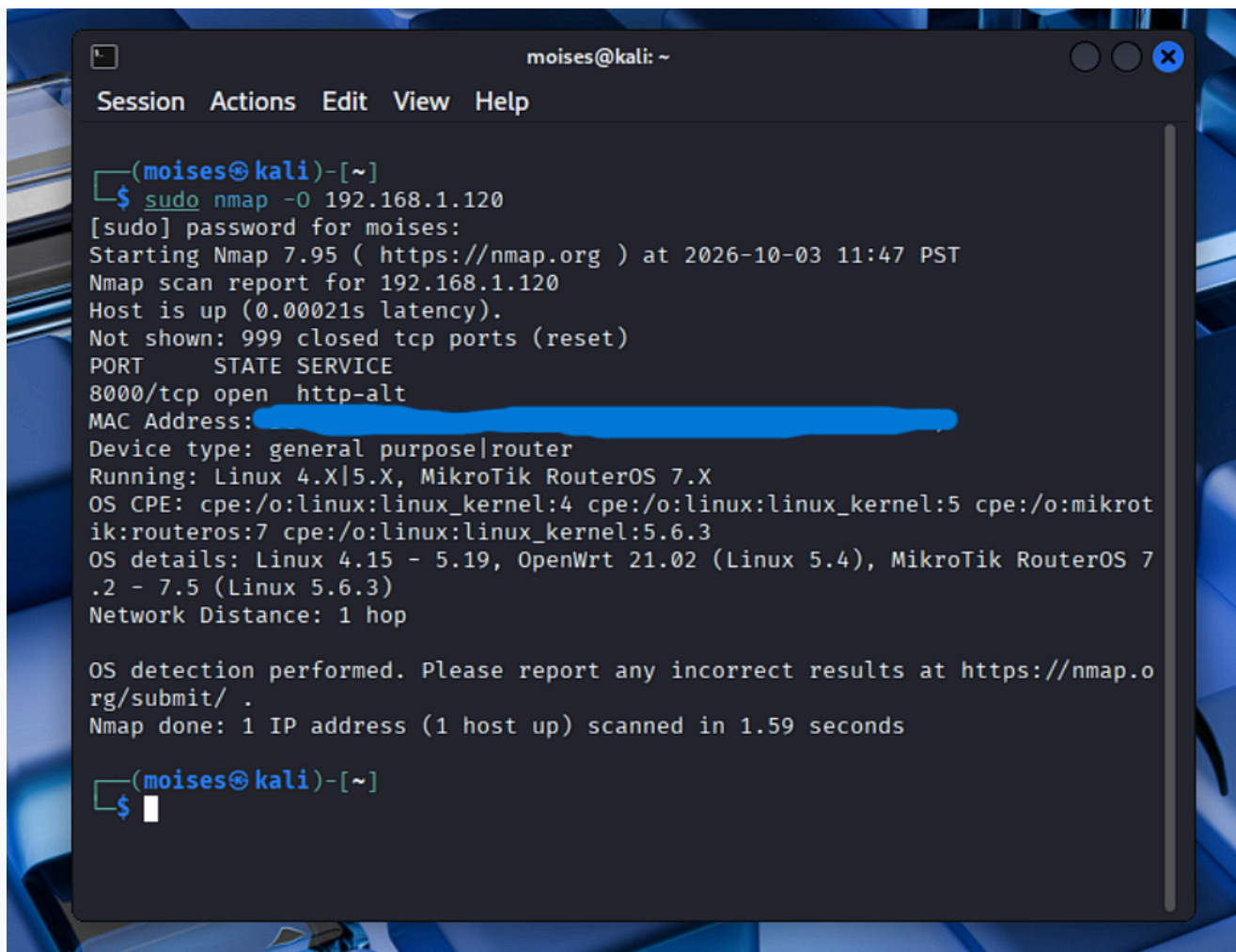
Run an authorized OS/platform identification scan against the Ubuntu Server target. Record the exact command and save the output.

COMMAND USED

```
sudo nmap -O 192.168.1.120
```

EVIDENCE 07 — OS Identification Result

Paste a screenshot of the result. Ensure the detected OS/platform and confidence/limitations are visible if Nmap provides them.



```
moises@kali: ~  
Session Actions Edit View Help  
  
(moises@kali)-[~]  
$ sudo nmap -O 192.168.1.120  
[sudo] password for moises:  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-03 11:47 PST  
Nmap scan report for 192.168.1.120  
Host is up (0.00021s latency).  
Not shown: 999 closed tcp ports (reset)  
PORT      STATE SERVICE  
8000/tcp  open  http-alt  
MAC Address:   
Device type: general purpose|router  
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X  
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3  
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.59 seconds  
  
(moises@kali)-[~]  
$
```

Figure 7.1. Nmap OS detection performed against the authorized Ubuntu Server target (192.168.1.120). Nmap returned a Linux/router-oriented fingerprint rather than identifying the installed Ubuntu Server

distribution, demonstrating a limitation of automated OS fingerprinting in the virtualized laboratory environment.

Step 2 — Validate the Observation

Compare the Nmap result with the operating system you actually installed on the target VM.

OBSERVATION	ACTUAL LAB CONFIGURATION	ASSESSMENT
Nmap OS/platform result	Ubuntu Server	Inconclusive / inaccurate
Reported platform	Ubuntu Server	Nmap reported Linux/router-oriented fingerprints
Detection limitation	Ubuntu Server running in a virtualized Proxmox environment	OS fingerprint should not be treated as definitive

Analysis

Nmap successfully identified the target as a reachable Linux-related system, but its OS fingerprint did not accurately identify the installed Ubuntu Server operating system. The result included Linux kernel ranges and router-oriented fingerprints such as OpenWrt and MikroTik RouterOS. Because the actual target configuration is known to be Ubuntu Server, the automated OS result is treated as inconclusive rather than as definitive identification. This demonstrates that OS fingerprinting should be validated against known system information and should not be treated as authoritative on its own.

08. Phase 6 — Security-Focused Service Exposure Assessment

This phase turns scan output into cybersecurity analysis. Do not label a service as a vulnerability solely because a port is open.

Step 1 — Select Relevant Services

From your service inventory, choose the services that are most relevant for security analysis. Prioritize services that are unnecessary, unexpectedly exposed, remotely accessible, or provide useful version information.

SERVICE	WHY EXPOSED?	EXPECTED?	RISK / SECURITY CONCERN	NEXT VALIDATION
HTTP / TCP 8000	Intentionally enabled for Nmap service enumeration	Yes — temporary lab service	Provides a remotely reachable HTTP service and therefore increases the target's network attack surface	Verify HTTP response/content and confirm service is disabled when no longer required

Step 2 — Analyze, Don't Overclaim

- ☐ Confirm the service belongs to the target VM.
- ☐ Determine whether the service is intentionally running.
- ☐ Record the detected product/version information if available.
- ☐ Use authoritative documentation to understand normal exposure and configuration.
- ☐ Only call something a vulnerability when there is evidence supporting that conclusion.
- ☐ Separate observation, interpretation, and recommended validation.

EVIDENCE 08 — HTTP Service Validation

Paste the terminal screenshot showing the HTTP response received from the authorized Ubuntu Server target using curl. The screenshot should show the HTTP status and server information.

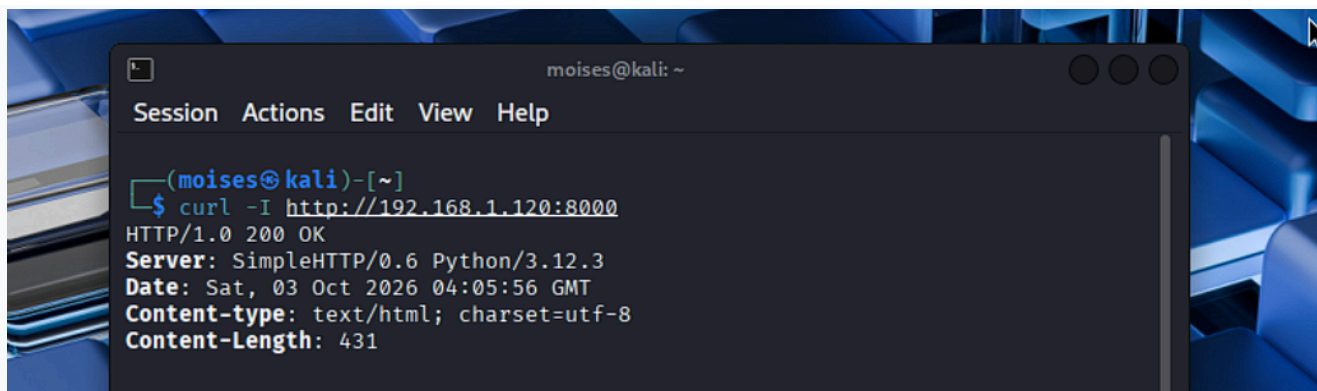


Figure 8.1. HTTP response validation for the authorized Ubuntu Server target on TCP/8000. The server returned HTTP 200 OK and identified itself as SimpleHTTP/0.6 running on Python 3.12.3, confirming that the service detected through Nmap was actively responding to HTTP requests.

Step 3 — Write One Detailed Finding

FINDING TITLE

F-01 — Temporary HTTP Service Exposure on TCP/8000

Observation: Nmap identified TCP/8000 as an open HTTP service on the authorized Ubuntu Server target. The service was identified as SimpleHTTPServer 0.6 running on Python 3.12.3.

Evidence: Nmap service enumeration identified TCP/8000 as HTTP. HTTP header validation using curl returned HTTP 200 OK and identified the server as SimpleHTTP/0.6 Python/3.12.3.

Security relevance: A remotely reachable HTTP service contributes to the target's network attack surface. In this laboratory, the service was intentionally enabled for testing and therefore does not by itself demonstrate a vulnerability.

Recommended validation / remediation: Confirm that the HTTP service is required for the intended laboratory activity. After testing is complete, terminate the temporary service and verify that TCP/8000 is no longer exposed.

Limitations: The assessment confirms service exposure and HTTP response behavior but does not establish that the service contains a vulnerability or was exploitable.

09. Phase 7 — Findings & Risk Summary

Use this page after all scan phases are complete. Keep findings evidence-based and specific to the lab.

ID	FINDING	EVIDENCE	IMPACT	RECOMMENDATION	STATUS
F-01	Temporary HTTP Service Exposure on TCP/8000	Evidence 06, Evidence 08	Increases the target's reachable network attack surface	Disable the temporary service after testing and verify TCP/8000 is no longer exposed	Validated

Evidence-to-Finding Traceability

F-01 is supported by Evidence 06 (Nmap service/version enumeration) and Evidence 08 (HTTP response validation). These evidence items establish that TCP/8000 was reachable and actively responding as an HTTP service.

10. Phase 8 — Optional Second Target

Once the Ubuntu Server workflow is complete, you may add a second intentionally configured target to make the assessment more realistic.

EVIDENCE 09 — Second Target in Proxmox

Paste a screenshot of the second target VM and the lab topology.

Key ↑	Value
cores	2
cpu	x86-64-v2-AES
ide2	local:iso/debian-13.5.0-amd64-netinst.iso,media=cdrom
memory	2048
name	[blurred]
net0	virtio,bridge=vmbr209801,firewall=1
nodename	[blurred]
numa	0
ostype	l26
pool	[blurred]
scsi0	local-lvm:10,iothread=on
scsihw	virtio-scsi-single
sockets	1
vmid	[blurred]

☐ Start after created

Advanced ☐ Back Finish

Figure 9.1. Second Target VM configured in the school-provided Proxmox environment, using Debian 13 Trixie with a 2-core CPU, 2 GB RAM, 10 GB storage, and the assigned laboratory network configuration.

<i>Configuration</i>	<i>Value</i>
<i>Target</i>	<i>Debian 13 Trixie</i>
<i>IP Address</i>	<i>192.168.1.141</i>
<i>CPU</i>	<i>2 cores</i>
<i>Memory</i>	<i>2 GB</i>
<i>Storage</i>	<i>10 GB</i>
<i>Network</i>	<i>VirtIO</i>
<i>Network Bridge</i>	<i>vibr209801</i>
<i>Network Assignment</i>	<i>DHCP</i>

EVIDENCE 10 — Target 2 Host Discovery

Insert your `nmap -sn 192.168.1.141` screenshot.

```
(moises@kali)-[~]
$ nmap -sn 192.168.1.141
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-03 23:31 PST
Nmap scan report for 192.168.1.141
Host is up (0.00022s latency).
MAC Address: [REDACTED]
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
(moises@kali)-[~]
```

Figure 10.1. Nmap host discovery of the authorized Debian 13 Target 2 (192.168.1.141), confirming that the laboratory VM was reachable and active from the Nmap assessment host.

EVIDENCE 11 — Target 2 Service Enumeration

```
(moises@kali)-[~]
$ nmap -sn 192.168.1.141
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-03 23:31 PST
Nmap scan report for 192.168.1.141
Host is up (0.00022s latency).
MAC Address: [REDACTED]
Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
(moises@kali)-[~]
```

Figure 11.1. Nmap service and version enumeration of the authorized Debian 13 Target 2 (192.168.1.141), identifying TCP port 22 as an SSH service running OpenSSH 10.0p2 on Debian.

EVIDENCE 12 — Target 2 OS Detection

```

(moises@kali)~$ sudo nmap -O 192.168.1.141
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-03 23:35 PST
Nmap scan report for 192.168.1.141
Host is up (0.00021s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: XXXXXXXXXX
No exact OS matches for host (if you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=10/3%OT=22%CT=1%CU=31230%PV=Y%D=1%DC=0%G=Y%M=BC2411%T
OS:M=6AC120C9%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10D%TI=Z%CI=Z%II=I
OS:%TS=21)SEQ(SP=105%GCD=1%ISR=10D%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=106%GCD=1%IS
OS:R=107%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=8%GCD=1%ISR=108%TI=Z%CI=Z%II=I%TS=21)
OS:SEQ(SP=8%GCD=1%ISR=105%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW7%O2=M5B4S
OS:T11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN(W1=
OS:FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=
OS:M5B4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+F=AS%RD=0%Q=)T2(R=N)T3(R=N)
OS:T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S
OS:+%F=AR%O=0%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=0%RD=0%Q=)T7(R=Y%DF=
OS:Y%T=40%W=0%S=Z%A=S+F=AR%O=0%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G
OS:%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.58 seconds

```

Figure 12.1. Nmap OS detection performed against the authorized Debian 13 Target 2. Nmap generated a TCP/IP fingerprint but did not identify an exact operating system match.

Comparison

ATTRIBUTE	TARGET 1 -Ubuntu Server	TARGET 2 - Debian 13	OBSERVATION
IP Address	192.168.1.120	192.168.1.141	Both targets were assigned distinct IP addresses within the laboratory network.
Operating System	Ubuntu Server	Debian 13 Trixie	Both are Linux-based server operating systems used as authorized laboratory targets.
Host Discovery	Host is up	Host is up	Nmap successfully identified both targets as reachable.
Exposed Service	TCP/8000	TCP/22	The targets presented different network services for assessment.

Analysis

The two authorized laboratory targets demonstrated different service exposure profiles. Target 1 exposed TCP/8000 as an intentionally enabled HTTP service, while Target 2 exposed TCP/22 as an SSH service. Nmap successfully identified the respective services and versions. However, OS fingerprinting did not produce exact operating system matches for either target, demonstrating a limitation of automated OS detection in the virtualized laboratory environment.

11. Evidence Capture Guide

You do not need dozens of screenshots. Capture a small number of strong, readable evidence items that prove the project was actually performed.

EVIDENCE	WHAT IT SHOULD PROVE	WHERE TO CAPTURE
01	Proxmox topology / VM	Proxmox UI
02	Target VM running	Proxmox UI
03	Connectivity	Terminal
04	Network boundary	Proxmox
05	Host discovery	Nmap terminal
06	Service/version enumeration	Nmap terminal
07	OS identification	Nmap terminal
08	Final findings	Report/table

12. Documentation Templates

A. Project README — Copy This Structure

Project Title

Nmap Network Reconnaissance & Service Exposure Assessment

Overview

This project documents an authorized network reconnaissance and service exposure assessment conducted within a controlled Proxmox-based cybersecurity laboratory environment.

The assessment used **Nmap** from a Kali Linux assessment VM to perform host discovery, service and version enumeration, OS fingerprinting, and validation of exposed network services against two intentionally created Linux virtual machines.

The project demonstrates how network reconnaissance results can be collected, validated, documented, and translated into security-relevant observations.

Scope

The assessment was restricted to two laboratory virtual machines intentionally created for this project:

Target	Operating System	IP Address
Target 1	Ubuntu Server	192.168.1.120
Target 2	Debian 13 Trixie	192.168.1.141

The Nmap assessment host was the Kali Linux VM at:

192.168.1.139

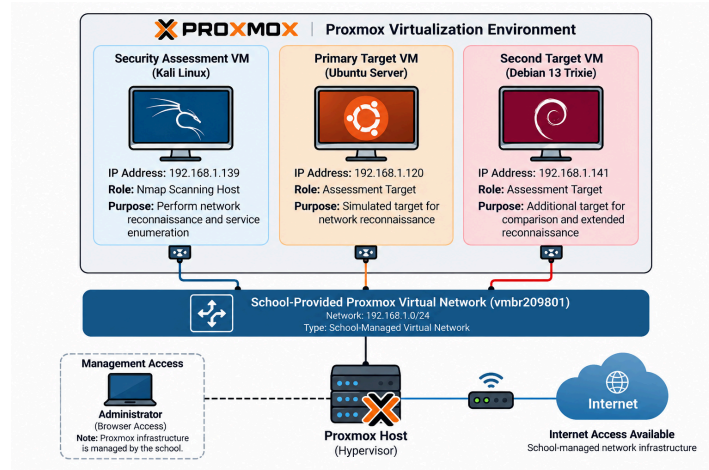
All scanning activities were limited to these authorized targets. The broader shared network was **not intentionally scanned**.

Objectives

1. Perform authorized host discovery against the assigned laboratory targets.
2. Identify exposed TCP services using Nmap.
3. Determine service and software versions where detectable.
4. Perform OS fingerprinting and document the accuracy or limitations of automated identification.
5. Validate identified services using appropriate network-level checks.
6. Compare the network service exposure of two Linux-based laboratory targets.
7. Document findings and recommendations based only on validated evidence.
8. Maintain an organized evidence trail suitable for cybersecurity portfolio documentation.

Lab Architecture

The virtual machines were deployed within the school-provided Proxmox environment using the assigned virtual network `vmb209801`. The laboratory network provides internet connectivity through school-managed infrastructure. No modifications were made to the underlying Proxmox network configuration or shared infrastructure.



Note: The Proxmox network infrastructure was school-provided. No bridge, physical network interface, or shared network infrastructure was modified as part of this project.

Laboratory Configuration

Assessment VM

Configuration	Value
Platform	Kali Linux
Role	Nmap assessment host
IP Address	192.168.1.139
Primary Tool	Nmap 7.95

Target 1 — Ubuntu Server

Configuration	Value
OS	Ubuntu Server
IP Address	192.168.1.120
Network Assignment	DHCP
Assessed Service	TCP/8000
Service	HTTP
Software	SimpleHTTPServer 0.6
Runtime	Python 3.12.3

The HTTP service was intentionally enabled for controlled service enumeration and validation.

Target 2 — Debian 13

Configuration	Value
OS	Debian 13 Trixie
IP Address	192.168.1.141
CPU	2 cores
RAM	2 GB
Storage	10 GB
Network Model	VirtIO
Network Assignment	DHCP
Assessed Service	TCP/22
Service	SSH
Software	OpenSSH 10.0p2 Debian 7+deb13u4

Tools & Technologies

- **Proxmox VE** — virtual laboratory infrastructure
- **Kali Linux** — assessment/scanning host
- **Nmap 7.95** — network reconnaissance and service enumeration
- **Ubuntu Server** — Target 1
- **Debian 13 Trixie** — Target 2
- **Python 3.12.3 SimpleHTTPServer** — intentionally enabled HTTP service on Target 1
- **OpenSSH 10.0p2** — SSH service on Target 2
- **cURL** — HTTP service validation

Methodology

Phase 1 — Laboratory Preparation

Created and configured the authorized virtual machines within the school-provided Proxmox environment.

Phase 2 — Network Verification

Verified IP addressing and connectivity between the assessment VM and the authorized targets.

Phase 3 — Host Discovery

Used Nmap host discovery to confirm that each authorized target was reachable.

Example:

```
nmap -sn 192.168.1.120
```

```
nmap -sn 192.168.1.141
```

Phase 4 — Service & Version Enumeration

Used Nmap service detection to identify exposed services and software versions.

Examples:

```
nmap -sV -p 8000 192.168.1.120
```

```
nmap -sV 192.168.1.141
```

Phase 5 — OS Fingerprinting

Used Nmap OS detection to attempt automated operating system identification.

```
sudo nmap -O 192.168.1.120
```

```
sudo nmap -O 192.168.1.141
```

Phase 6 — Service Validation

Validated the HTTP service on Target 1 using cURL:

```
curl -I http://192.168.1.120:8000
```

The service returned **HTTP/1.0 200 OK**, confirming that the detected HTTP service was actively responding.

Phase 7 — Findings Analysis

Reviewed the collected reconnaissance and validation results and documented only observations supported by the captured evidence.

Phase 8 — Comparative Assessment

Compared the service exposure of the Ubuntu and Debian targets to demonstrate differences in network-facing services.

Findings

F-01 — Temporary HTTP Service Exposure on TCP/8000

Target: Ubuntu Server (192.168.1.120)

Nmap identified TCP/8000 as an open HTTP service running **SimpleHTTPServer 0.6 on Python 3.12.3**. Subsequent HTTP validation using cURL returned **HTTP 200 OK**.

The service was intentionally enabled within the laboratory for assessment purposes. Therefore, the observation demonstrates **service exposure and network attack surface**, but does not by itself establish a vulnerability or exploitability.

Target 2 — SSH Service Exposure

The Debian 13 target (**192.168.1.141**) exposed TCP/22 as an SSH service. Nmap identified the service as **OpenSSH 10.0p2 Debian 7+deb13u4**.

The presence of an SSH service represents an exposed network service; the scan alone does not establish that the service is vulnerable.

OS Fingerprinting Limitation

Nmap did not produce an exact OS match for either target. The actual operating systems were known from the laboratory configuration, demonstrating that automated OS fingerprinting may be inconclusive in a virtualized environment.

Target Comparison

Attribute	Ubuntu Server	Debian 13
IP Address	192.168.1.120	192.168.1.141
Host Status	Host is up	Host is up
Exposed Port	8000/tcp	22/tcp
Service	HTTP	SSH
Detected Version	SimpleHTTPServer 0.6 / Python 3.12.3	OpenSSH 10.0p2 Debian 7+deb13u4
OS Detection	Inconclusive	Inconclusive
Observation	Temporary HTTP service intentionally exposed for testing	SSH service available on the Debian target

Recommendations

1. Disable temporary laboratory services after testing is complete.
2. Verify that TCP/8000 is no longer exposed after the HTTP testing activity.
3. Review whether SSH access on TCP/22 is required for the intended target configuration.
4. Restrict network-facing services to those required for the system's intended purpose.
5. Perform service and version enumeration only against authorized systems.
6. Preserve scan results and supporting evidence for reproducibility and documentation.

Limitations

- The assessment was restricted to the two intentionally created laboratory targets.
- The broader shared Proxmox network was not intentionally scanned.
- The assessment focused on reconnaissance, service exposure, and validation rather than exploitation.
- No credential attacks or unauthorized access attempts were performed.
- Nmap OS fingerprinting did not produce exact OS matches for either target.
- An exposed service was not automatically treated as a vulnerability.
- Findings were limited to observations supported by the collected evidence.

Evidence

The project evidence is organized according to the assessment workflow:

evidence/

- |— 01-proxmox-topology/
- |— 02-target-running/
- |— 03-connectivity/
- |— 04-network-boundary/
- |— 05-host-discovery/
- |— 06-service-enumeration/
- |— 07-os-detection/
- |— 08-findings/
- |— 09-second-target/
- |— 10-target2-host-discovery/
- |— 11-target2-service-enumeration/
- |— 12-target2-os-detection/

Each evidence item contains the corresponding screenshot or documentation supporting the assessment result.

Conclusion

This project demonstrated a structured and authorized approach to network reconnaissance within a controlled virtualized cybersecurity laboratory. Nmap was used to identify reachable hosts, enumerate network services and versions, perform OS fingerprinting, and support service exposure analysis.

The comparison between the Ubuntu Server and Debian 13 targets demonstrated that different systems can present different network-facing services. The project also highlighted the importance of validating automated scan results, documenting limitations, and distinguishing observable service exposure from confirmed vulnerabilities.

13. Technical Report Template

Executive Summary

This laboratory project documents an authorized network reconnaissance and service exposure assessment conducted within a controlled Proxmox-based cybersecurity environment. The assessment involved a Kali Linux Nmap host and two intentionally created Linux virtual machines: an Ubuntu Server target and a Debian 13 Trixie target. Nmap was used to perform host discovery, service and version enumeration, and OS fingerprinting, while cURL was used to validate the HTTP service identified on the Ubuntu target.

The assessment identified different service exposure profiles between the two targets. The Ubuntu Server exposed an intentionally enabled HTTP service on TCP/8000, while the Debian target exposed SSH on TCP/22. OS fingerprinting did not produce exact matches for either target. The project demonstrates authorized reconnaissance, evidence collection, service validation, and security-focused analysis within a controlled laboratory environment.

A. Environment

The laboratory was implemented using a **school-provided Proxmox environment** containing an assessment VM and two intentionally created Linux target VMs.

System	Operating System	IP Address	Role
Assessment VM	Kali Linux	192.168.1.139	Nmap scanning and assessment
Target 1	Ubuntu Server	192.168.1.120	Authorized assessment target
Target 2	Debian 13 Trixie	192.168.1.141	Authorized assessment target

The Proxmox network infrastructure was provided by the school. No shared network infrastructure, bridge configuration, or physical networking components were modified as part of the assessment.

All scanning activities were restricted to the specifically assigned laboratory targets.

B. Methodology

The assessment followed a structured reconnaissance workflow:

1. **Network verification** — verified addressing and connectivity between the assessment VM and authorized targets.

2. **Host discovery** — used Nmap to confirm target availability.
3. **Service and version enumeration** — identified exposed TCP services and detectable software versions.
4. **OS identification** — used Nmap OS fingerprinting to evaluate automated operating system identification.
5. **Service validation** — used cURL to validate the HTTP service identified on Target 1.
6. **Comparative assessment** — compared the service exposure profiles of the Ubuntu and Debian targets.
7. **Findings analysis** — documented observations based on collected evidence without treating service exposure alone as proof of vulnerability.

C. Results

Target 1 — Ubuntu Server

Test	Result
Host Discovery	Host is up
Service	HTTP
Port	8000/tcp
Version	SimpleHTTPServer 0.6 / Python 3.12.3
HTTP Validation	HTTP 200 OK
OS Detection	Inconclusive

Target 2 — Debian 13

Test	Result
Host Discovery	Host is up
Service	SSH
Port	22/tcp
Version	OpenSSH 10.0p2 Debian 7+deb13u4
OS Detection	Inconclusive

Comparative Observation

The two authorized targets presented different network service profiles. Ubuntu exposed an intentionally enabled HTTP service on TCP/8000, while Debian exposed SSH on TCP/22. Nmap successfully identified the services and versions, while automated OS fingerprinting did not produce exact matches for either target.

D. Findings

F-01 — Temporary HTTP Service Exposure on TCP/8000

Observation:

Nmap identified TCP/8000 as an open HTTP service on the authorized Ubuntu Server target. The service was identified as SimpleHTTPServer 0.6 running on Python 3.12.3.

Evidence:

Nmap service/version enumeration and cURL HTTP response validation.

Security Relevance:

An actively responding network service contributes to the target's reachable network attack surface. In this laboratory, the service was intentionally enabled for controlled assessment and therefore does not by itself demonstrate a vulnerability.

Recommendation:

Disable the temporary HTTP service after testing and verify that TCP/8000 is no longer exposed.

Limitation:

The assessment confirmed service exposure and HTTP response behavior but did not test for exploitation or establish vulnerability of the service.

Target 2 — SSH Service Exposure on TCP/22

Observation:

Nmap identified TCP/22 as an open SSH service on the authorized Debian 13 target and identified the service as OpenSSH 10.0p2 Debian 7+deb13u4.

Evidence:

Nmap service/version enumeration against 192.168.1.141.

Security Relevance:

SSH represents a network-accessible administrative service and therefore contributes to the target's exposed service surface.

Recommendation:

Confirm that SSH access is required for the intended laboratory configuration and restrict access appropriately where applicable.

Limitation:

The assessment identified service exposure but did not perform credential attacks, exploitation, or vulnerability validation.

OS Fingerprinting Observation

Observation:

Nmap OS detection did not produce an exact operating system match for either authorized target.

Evidence:

Nmap OS detection results for 192.168.1.120 and 192.168.1.141.

Security Relevance:

This demonstrates that automated OS fingerprinting can be inconclusive in virtualized laboratory environments.

Limitation:

The actual operating systems were known from the laboratory configuration, so the Nmap results were treated as inconclusive rather than as definitive OS identification.

E. Recommendations

1. Disable temporary services after completing laboratory testing.
2. Verify that intentionally exposed services are no longer reachable when they are no longer required.
3. Review whether SSH access is necessary for the target's intended purpose.
4. Limit network-facing services to those required for system operation.
5. Validate reconnaissance results using additional evidence where appropriate.
6. Maintain clear authorization boundaries when performing network scanning.

F. Limitations

The assessment was limited to the authorized laboratory targets created for this project. It did not include:

- Exploitation of identified services
- Credential attacks
- Password cracking
- Unauthorized access attempts
- Scanning of external systems
- Scanning of unrelated systems within the shared environment
- Full vulnerability assessment of the target operating systems
- Production network testing

OS fingerprinting was also inconclusive for both targets.

G. Conclusion

This laboratory demonstrated a structured approach to authorized network reconnaissance and service exposure assessment using Nmap within a controlled Proxmox environment. The assessment successfully identified reachable hosts, exposed services, service versions, and differences in network-facing services between two Linux targets.

The project also demonstrated the importance of interpreting reconnaissance results carefully. An exposed service was documented as a service exposure rather than automatically classified as a vulnerability, while inconclusive OS fingerprinting results were reported as limitations. These practices support accurate

evidence-based cybersecurity assessment and demonstrate practical skills relevant to network security and SOC/Blue Team activities.

13. Project Completion Record

Complete this page after the laboratory and documentation are finished.

Project Status	In Progress
Date Started	October 3, 2026
Date Completed	Pending final documentation
GitHub Repository	To be added after repository creation
Primary Target	Ubuntu Server — 192.168.1.120
Scanner VM	Kali Linux — 192.168.1.139
Second Target	Debian 13 Trixie — 192.168.1.141
Final Evidence Count	12

Authenticity & Scope Statement

I confirm that the technical activities documented in this project were performed in an authorized and controlled laboratory environment created for cybersecurity learning and portfolio development. The Nmap assessments were limited to intentionally configured virtual machines within the laboratory scope. Findings are based on documented observations and are not presented as evidence of compromise or vulnerabilities in any external or production system.

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Date: October 4, 2026